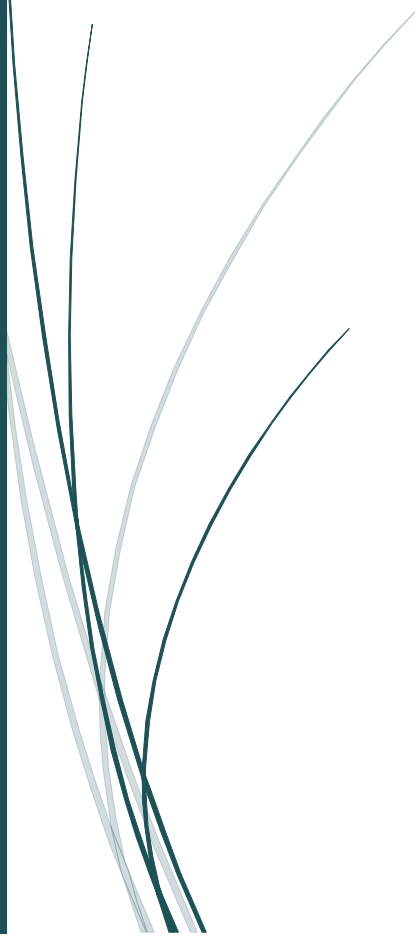
9/1/2026

N + - Network Fundamentals & Building Networks.

Switching Basics.



YAGNIK GHETIYA

TASK 1.

❖ Open Cisco Packet Tracer or any network simulator, create a simple network with one switch and two PCs, and use the 'show mac address-table' command to observe how the MAC address table is initially empty.

- **Observing an Empty MAC Address Table:**

  Network Setup: One switch and two PCs were connected in Cisco Packet Tracer.

  Check MAC Address Table: The following command was entered on the switch:

  Switch> enable
  Switch# show mac address-table

  Observation: Initially, the MAC address table was empty because the switch had not yet learned the PCs' MAC addresses.

  Result: The switch was ready to learn MAC addresses when network traffic started flowing.

TASK 2.

❖ Send a ping from PC1 to PC2 in your simulated network, then immediately run 'show mac address-table' on the switch and note how the MAC addresses of both PCs appear in the table, Try pinging in both directions to see how the table updates.

- **Observing MAC Address Learning:**

  Ping Test: From PC1, ping PC2, then repeat the test from PC2 to PC1.

  ping 192.168.1.2

  Check MAC Address Table: On the switch, run**:**

  Switch# show mac address-table

  Observation: The switch learns the MAC addresses of both PCs and records them against their connected switch ports.

  Result: The MAC address table updates as traffic passes through the switch in both directions.

TASK 3.

❖ Add a third PC to your network, connect it to the switch, and ping from PC3 to PC1. Observe and record how the MAC address table changes after this communication.

- **Observing MAC Address Table Updates:**

Network Setup: A third PC (PC3) was connected to the switch.
Ping Test: From PC3, ping PC1.

ping 192.168.1.1

Check MAC Address Table: Run this command on the switch.

Switch# show mac address-table

Observation: The switch learns PC3's MAC address and associates it with PC3's connected port. The table now contains the MAC addresses of all three PCs.

Result: The MAC address table updated automatically after PC3 communicated with PC1.

TASK 4.

- **Suppose you see the following MAC address table on a switch: Port 1 - 00:11:22:33:44:55, Port 2 - 66:77:88:99:AA:BB. If a frame arrives at the switch on Port 1 with destination MAC 66:77:88:99:AA:BB, explain step-by-step how the switch will forward this frame.**

- **How a Switch Forwards a Frame:**

  - **Frame Reception:** The switch receives the frame on Port 1.
  - **MAC Address Lookup:** It checks the destination MAC address 66:77:88:99:AA:BB in its MAC address table.
  - **Port Identification:** The table shows that this MAC address belongs to Port 2.
  - **Frame Forwarding:** The switch forwards the frame only through Port 2, not to all other ports.

Result: The frame reaches its intended destination through Port 2.

TASK 5.

❖ **Draw a diagram showing how a switch learns and uses MAC addresses to forward frames in a network with 4 devices, and write a short explanation (3-4 sentences) describing the process in your own words, Do not copy definitions from the textbook; use your own understanding and examples.**

➢ **MAC Address Learning and Frame Forwarding:**

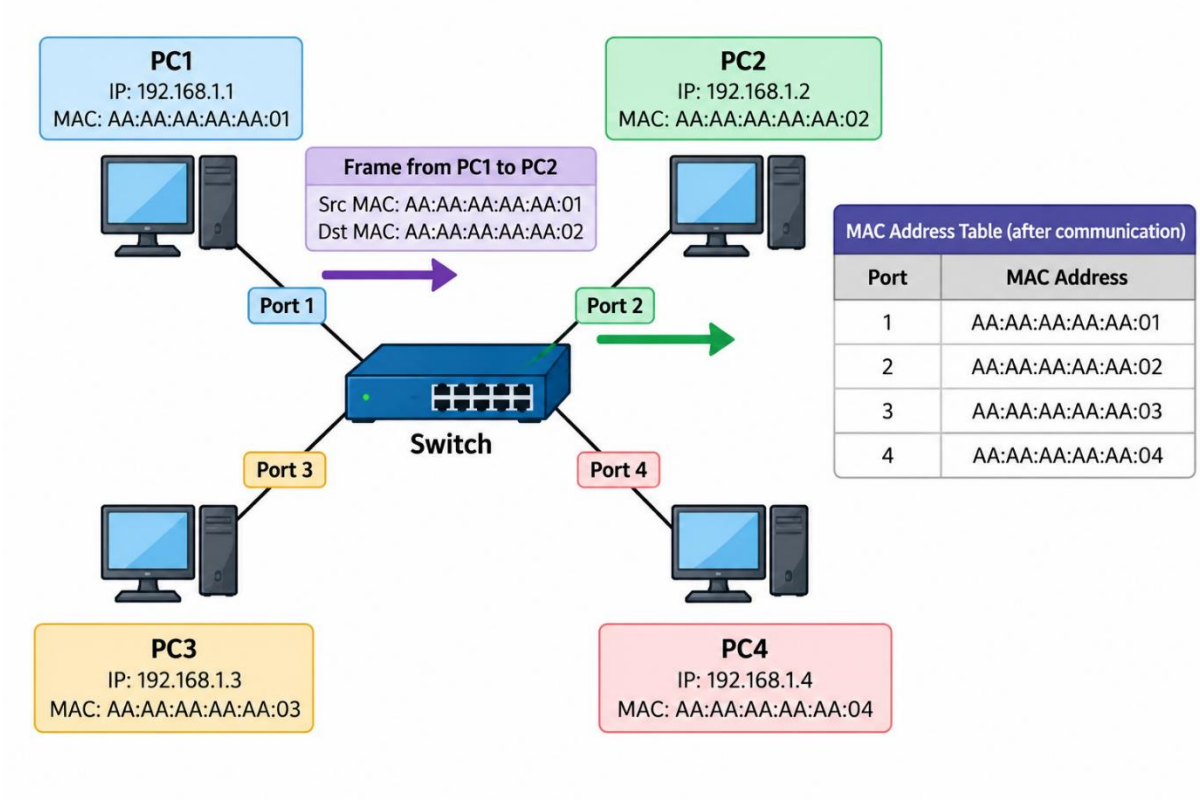


Figure : MAC Address Learning and Frame Forwarding in a Four-Device Network.

Explanation:

When a device sends a frame, the switch learns its source MAC address and saves it with the connected port. When another frame arrives, the switch checks the destination MAC address in its table. If the address is known, it forwards the frame only to the correct port. If the address is unknown, it sends the frame to all other ports except the one it arrived on.